Practice workbook · Where a line meets a curve (simultaneous equations)

Solving one linear and one non-linear equation together, either stated as algebra or as a line and a curve (or two curves) on a diagram.

19 question parts: 1 worked-example retry and 18 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-13

1. Question 3 · part d · 4 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part d only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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1. Question 3 · part d · 4 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

5 Use algebra to solve the equations

$$xy = 36$$

$$xy + x + 2y = 53$$

(6)

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2. Question 5 · part whole · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 6 marks)

2. Question 5 · part whole · 6 marks · Working and answer

Working and answer

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3. Question 2 · part a · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 One solution to the following simultaneous equations

$$y = px + 9$$
$$6x^2 - xy = 5$$

is $\left(-\frac{1}{2}, q\right)$, where p is an integer and q is a prime number.

(a) Find the value of p and the value of q

(4)

(b) Hence find the other solution to the equations.

(4)

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3. Question 2 · part a · 4 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 One solution to the following simultaneous equations

$$y = px + 9$$
$$6x^2 - xy = 5$$

is $\left(-\frac{1}{2}, q\right)$, where p is an integer and q is a prime number.

(a) Find the value of p and the value of q

(4)

(b) Hence find the other solution to the equations.

(4)

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4. Question 2 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 6 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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16

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5. Question 6 · part c · 5 marks · Working and answer

Working and answer

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6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

2

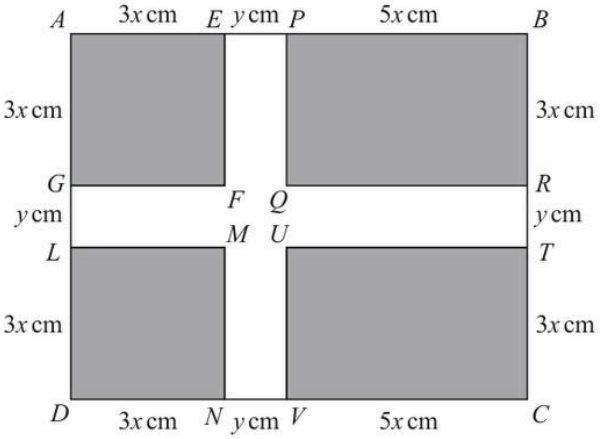


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the design for a flag consisting of a white cross on a grey background.

AEFG and *DLMN* are squares with sides of length $3x$ cm.
BPQR and *CTUV* are rectangles with sides of length $5x$ cm and $3x$ cm.
The width of the cross is y cm.

The total area of the flag is H cm^2

(a) Write down an expression, in terms of x and y, for H.

(1)

Given that the area of the cross is K cm^2

(b) show that $K = 14xy + y^2$

(3)

The total area of the flag is to be 3432 cm^2 and the area of the cross is to be 1080 cm^2

(c) Find the value of x and the value of y

(5)

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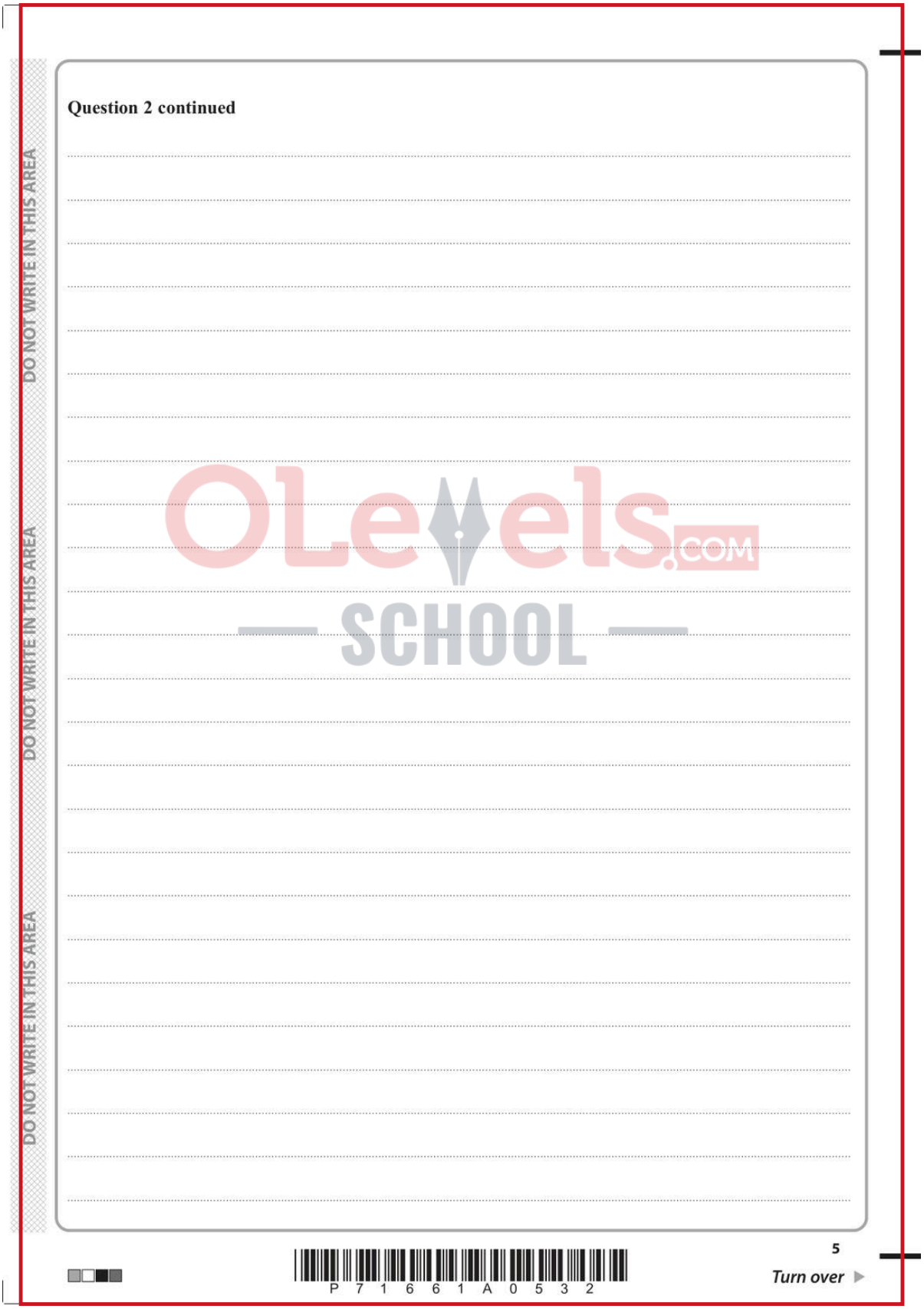
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6. Question 2 · part c · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.



Question 2 continued

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5

Turn over

6. Question 2 · part c · 5 marks

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Further practice · Attempt part c only.

Question 2 continued

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6. Question 2 · part c · 5 marks

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Further practice · Attempt part c only.

Question 2 continued

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(Total for Question 2 is 9 marks)

6. Question 2 · part c · 5 marks · Working and answer

Working and answer

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7. Question 2 · part b · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

2 (a) Express $x^2 + 4x - 8$ in the form $(x + a)^2 + b$ where a and b are constants whose values are to be found.

(2)

(b) Use algebra to solve the simultaneous equations

$$y = x^2 + 4x - 8$$
$$y = 2x + 7$$

(5)

The curve C has equation $y = x^2 + 4x - 8$
The straight line L has equation $y = 2x + 7$

Using the same axes and the results of part (a) and part (b),

(c) sketch C and L, showing clearly the coordinates of the turning point of C and the coordinates of the points of intersection of C and L.

(4)

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7. Question 2 · part b · 5 marks · Working and answer

Working and answer

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8. Question 10 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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8. Question 10 · part c · 4 marks · Working and answer

Working and answer

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9. Question 7 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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9. Question 7 · part c · 4 marks · Working and answer

Working and answer

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10. Question 5 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

5 Given that k is a non-zero constant

curve C has equation $kx^2 - xy + (k+1)x = 1$

straight line l has equation $y = \frac{k}{2}x + 1$

The point A is the only point that lies on both C and l.

(a) Find the value of k

(6)

(b) Hence, find the coordinates of A.

(2)

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10. Question 5 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

10. Question 5 · part b · 2 marks · Working and answer

Working and answer

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11. Question 10 · part a · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

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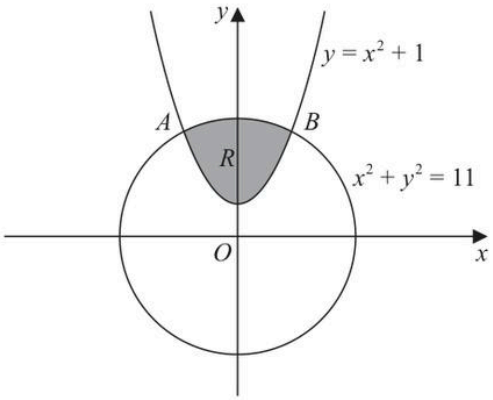


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Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = x^2 + 1$ and the curve with equation $x^2 + y^2 = 11$

The two curves intersect at the point A and at the point B.

(a) Find the x coordinate of the point A and the x coordinate of the point B.

(4)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume, to 2 decimal places, of the solid generated.

(5)

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11. Question 10 · part a · 4 marks · Working and answer

Working and answer

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12. Question 10 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

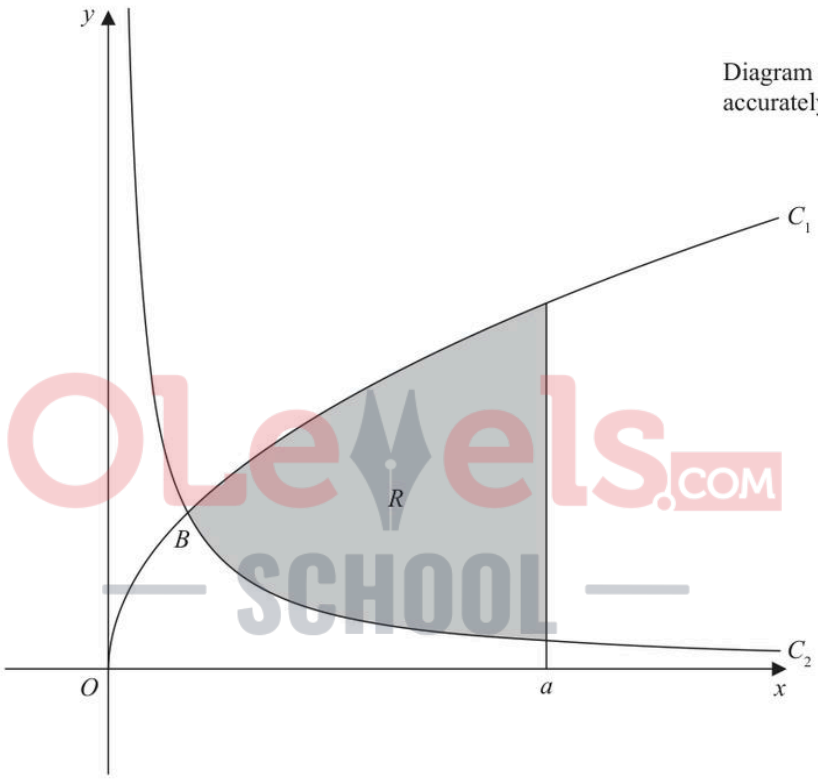


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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12. Question 10 · part c · 2 marks · Working and answer

Working and answer

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13. Question 5 · part a · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

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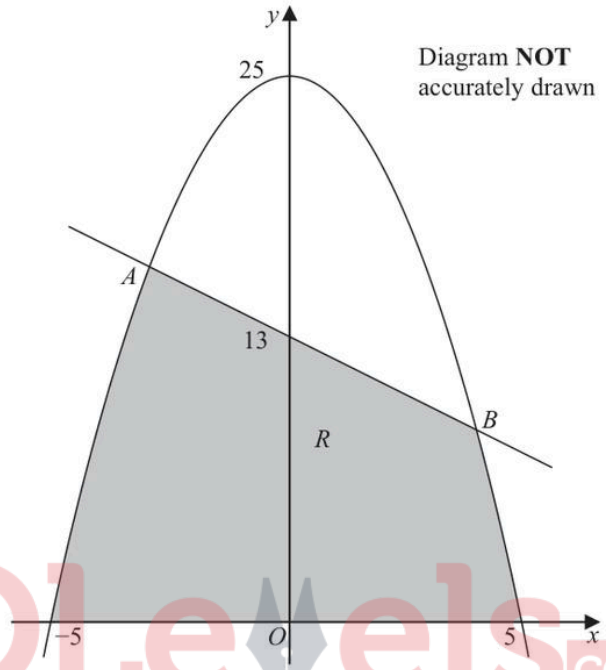


Figure 2

Figure 2 shows part of the curve with equation $y = 25 - x^2$ and part of the line with equation $y + x = 13$

The curve and the line intersect at the points A and B.

(a) Use algebra to find the coordinates of A and the coordinates of B.

(4)

The region R, shown shaded in Figure 2, is bounded by the curve, the straight line and the x-axis.

(b) Use algebraic integration to find the area of R.

(7)

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13. Question 5 · part a · 4 marks · Working and answer

Working and answer

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14. Question 11 · part c · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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14. Question 11 · part c · 3 marks · Working and answer

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15. Question 6 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

6

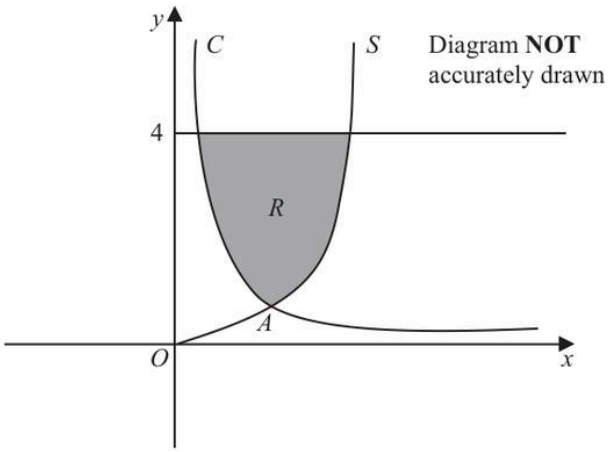


Figure 3

Figure 3 shows part of the curve C with equation $y = \frac{1}{4x}$, $x > 0$ and part of the curve S with equation $y = 2x^2$, $x \geqslant 0$

The curve C and the curve S intersect at the point A

(a) Find the coordinates of point A

(3)

The finite region R, shown shaded in Figure 3, bounded by the curve C, the curve S and the straight line $y = 4$ is rotated through 360° about the y-axis.

(b) Find, using algebraic integration, the exact volume of the solid formed.

(7)

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15. Question 6 · part a · 3 marks · Working and answer

Working and answer

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16. Question 2 · part a · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

2 (a) Use algebra to find the x coordinates of the points where the curve with equation $y = 3x^2 + 9x - 17$ intersects the line with equation $y = 3 - 2x$

(3)

(b) Hence, or otherwise, find the set of values of x for which $3 - 2x \leqslant 3x^2 + 9x - 17$

(2)

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16. Question 2 · part a · 3 marks · Working and answer

Working and answer

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17. Question 6 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

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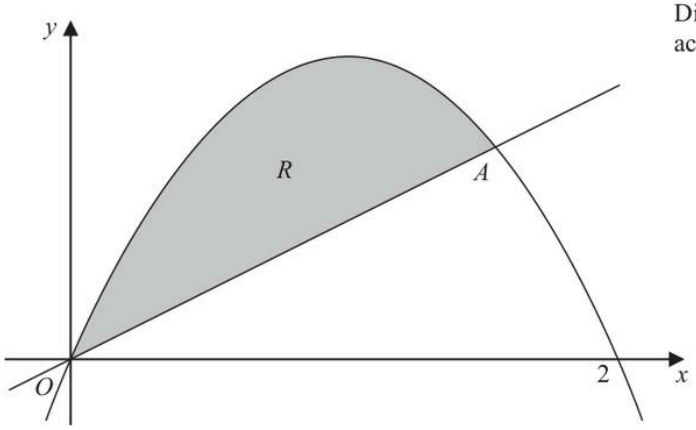


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = 2x - x^2$ and the line with equation $2y - x = 0$

The curve and the line intersect at the origin O and the point A.

(a) Show that the point A has coordinates $\left(\frac{3}{2}, \frac{3}{4}\right)$.

(2)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find, in terms of π, the volume of the solid formed.

(6)

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17. Question 6 · part a · 2 marks · Working and answer

Working and answer

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18. Question 4 · part a.i · 0 marks

2023 JUN · Paper 02

Further practice · Attempt part a.i only.

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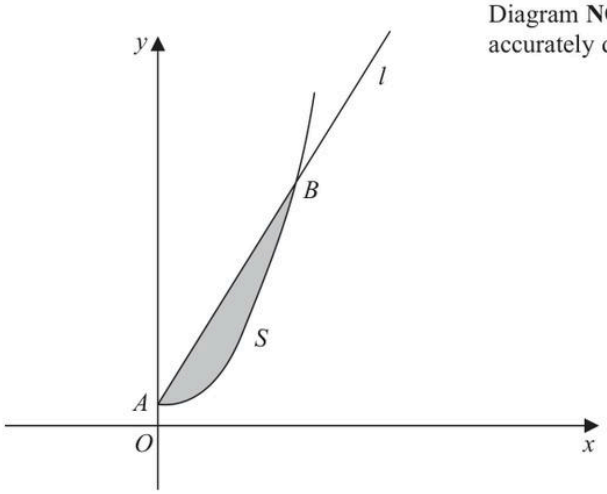


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Figure 2

The curve S with equation $y = \frac{x^2}{4} + 2$ where $x \geqslant 0$ and the line l with equation $2y - x - 4 = 0$ where $x \geqslant 0$ intersect at the points A and B, as shown in Figure 2.

(a) (i) Show that the coordinates of point A are $(0, 2)$

(ii) Find the coordinates of the point B

(4)

The finite region bounded by S and l, shown shaded in Figure 2, is rotated through 2π radians about the y-axis.

(b) Use algebraic integration to find the volume of the solid generated. Give your answer in terms of π

(4)

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18. Question 4 · part a.i · 0 marks · Working and answer

Working and answer

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19. Question 4 · part a.ii · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part a.ii only.

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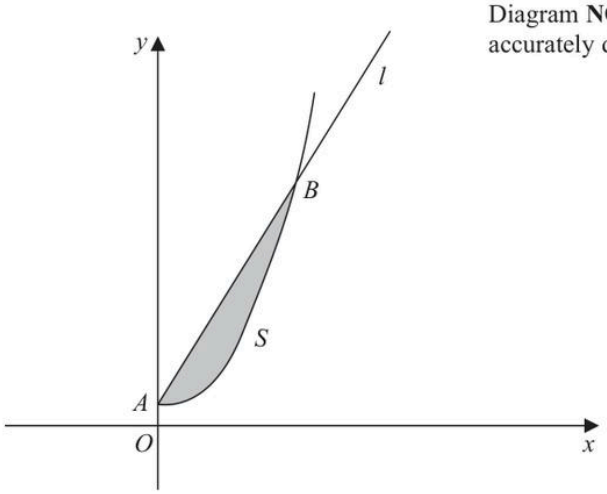


Diagram **NOT** accurately drawn

Figure 2

The curve S with equation $y = \frac{x^2}{4} + 2$ where $x \geqslant 0$ and the line l with equation $2y - x - 4 = 0$ where $x \geqslant 0$ intersect at the points A and B, as shown in Figure 2.

(a) (i) Show that the coordinates of point A are $(0, 2)$

(ii) Find the coordinates of the point B

(4)

The finite region bounded by S and l, shown shaded in Figure 2, is rotated through 2π radians about the y-axis.

(b) Use algebraic integration to find the volume of the solid generated. Give your answer in terms of π

(4)

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19. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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